synthesis and duplication of the genome) between the first and second meiotic divisions.

Meiosis II is similar to an ordinary mitosis, except that the chromosome number is 23 instead of 46; the chromatids of each of the 23 chromosomes separate, and one chromatid of each chromosome passes to each daughter cell (see Fig. 2-14). However, as mentioned earlier, because of crossing over in meiosis I, the chromosomes of the resulting gametes are not identical (see Fig. 2-15).

HUMAN GAMETOGENESIS AND FERTILIZATION

The cells in the germline that undergo meiosis, primary spermatocytes or primary oocytes, are derived from the zygote by a long series of mitoses before the onset of meiosis. Male and female gametes have different histories, marked by different patterns of gene expression that reflect their developmental origin as an XY or XX embryo. The human primordial germ cells are recognizable by the fourth week of development outside the embryo proper, in the endoderm of the yolk sac. From there, they migrate during the sixth week to the genital ridges and associate with somatic cells to form the primitive gonads, which soon differentiate into testes or ovaries, depending on the cells' sex chromosome constitution (XY or XX), as we examine in greater detail in Chapter 6. Both spermatogenesis and oogenesis require meiosis but have important differences in detail and timing that may have clinical and genetic consequences for the offspring. Female meiosis is initiated once, early during fetal life, in a limited number of cells. In contrast, male meiosis is initiated continuously in many cells from a dividing cell population throughout the adult life of a male.

In the female, successive stages of meiosis take place over several decades—in the fetal ovary before the female in question is even born, in the oocyte near the time of ovulation in the sexually mature female, and after fertilization of the egg that can become that female's offspring. Although postfertilization stages can be studied in vitro, access to the earlier stages is limited. Testicular material for the study of male meiosis is less difficult to obtain, inasmuch as testicular biopsy is included in the assessment of many men attending infertility clinics. Much remains to be learned about the cytogenetic, biochemical, and molecular mechanisms involved in normal meiosis and about the causes and consequences of meiotic irregularities.

Spermatogenesis

The stages of spermatogenesis are shown in Figure 2-16. The seminiferous tubules of the testes are lined with **spermatogonia,** which develop from the primordial

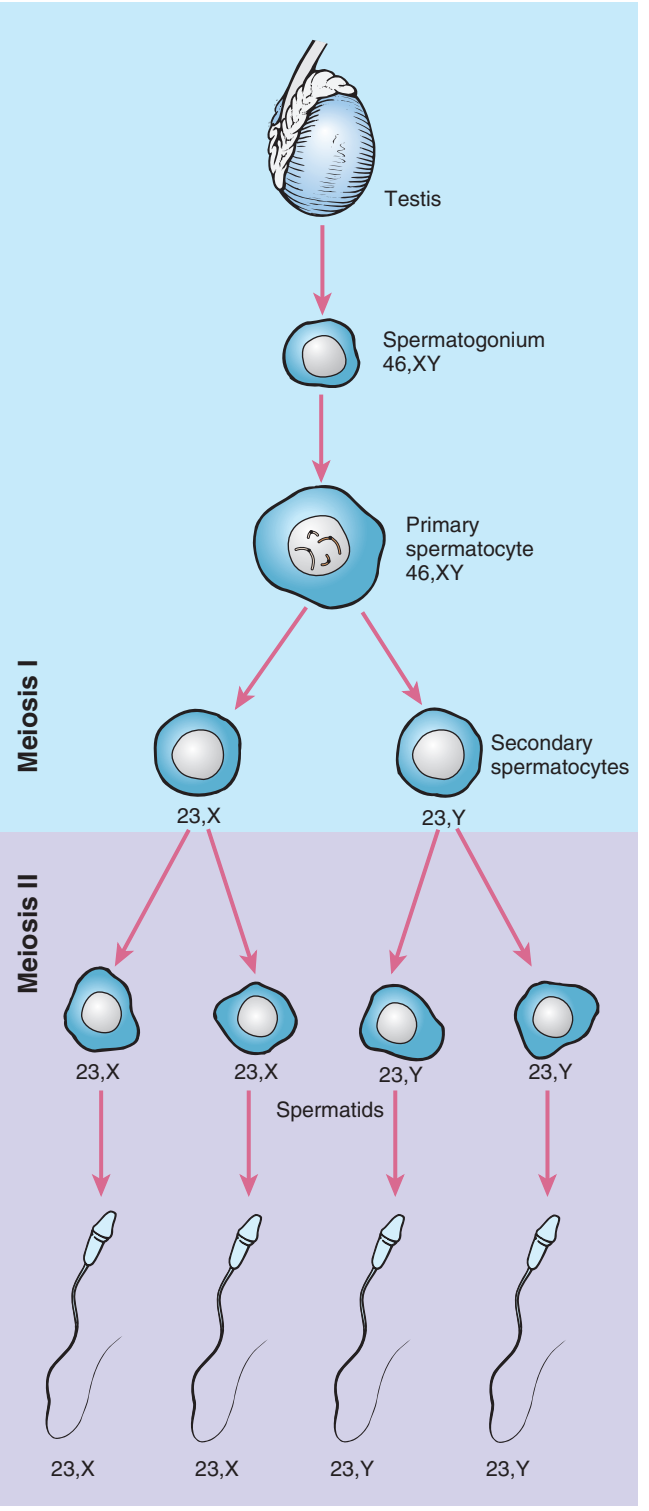


Figure 2-16 Human spermatogenesis in relation to the two meiotic divisions. The sequence of events begins at puberty and takes approximately 64 days to be completed. The chromosome number (46 or 23) and the sex chromosome constitution (X or Y) of each cell are shown. *See Sources & Acknowledgments.*